Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Excel Power Query and Power Pivot

TGS Ref No: TGS-2026064177

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version v5.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v4.0	16 August 2024	Previous release under course code TGS-2024048312.	Tertiary Infotech Academy
v5.0	12 August 2026	Re-issued for course code TGS-2026064177. Content realigned to the published three-topic outline; out-of-scope statistics material (hypothesis testing, t-tests, regression) removed. 19 hands-on labs added with starter workbooks. Slide references refreshed.	Tertiary Infotech Academy

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1. Course Overview

Course Title	Excel Power Query and Power Pivot
Course Reference Number	TGS-2026064177
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN: 201200696W)
Course Duration	2 days · 16 instructional hours (8 hours per day, excluding lunch)
Assessment Duration	2 hours (Written Assessment 1 hour + Practical Performance 1 hour)
Mode of Delivery	Classroom-based, instructor-led with hands-on practical labs
Class Size	Maximum 20 learners
TSC Title	Data and Statistical Analytics
TSC Code	EGS-BIN-4055-1.1
Software Required	Microsoft Excel for Windows (Microsoft 365, 2021, 2019 or 2016). Power Query is built in under Data > Get & Transform Data. Power Pivot is enabled from File > Options > Add-ins > COM Add-ins.

2. Learning Outcomes

By the end of this course, learners will be able to:

- LO1: Review data sets using Excel Power Query to uncover trends and patterns.
- LO2: Develop methods in Power Pivot to conduct complex analyses tailored to specific datasets.
- LO3: Apply advanced Power Query analytics to examine business issues.

3. Skills Framework Alignment

TSC Title: Data and Statistical Analytics · TSC Code: EGS-BIN-4055-1.1

Knowledge assessed:

- K1 Operations of data transformation techniques, e.g. shaping, cleansing, merging and aggregation
- K2 Conditions required for each transformation and join technique
- K3 Interpretation of the results from a query and a data model
- K4 Types of data analysis software and their capabilities

Abilities assessed:

- A1 Review data sets to uncover trends or patterns
- A2 Develop new methods to conduct analysis of large complex data sets specific to each issue
- A3 Facilitate the discussion on areas for application of data analytics to examine issues

4. Target Audience and Prerequisites

Target audience

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Data Analyst, Business Analyst, Financial Analyst, Marketing Analyst, Operations Analyst, Business Intelligence Developer, Report Developer, Data Consultant, Database Administrator, Systems Analyst.

Prerequisites

- Able to operate using computer functions
- Minimum 3 GCE 'O' Levels Passes including English, or WPL Level 5
- Minimum 1 year working experience
- Positive learning attitude and an enthusiastic learner

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5. Detailed Lesson Plan

Each training day runs 9:30am to 6:30pm and delivers exactly 8 instructional hours. The 1-hour lunch break is excluded from instructional time; tea breaks are taken within the sessions shown.

Day 1 — Getting, shaping and cleaning data with Power Query

Time	Dur.	Session / Topic	Content and Activities	Instructional Method	Slide Ref
9:30 – 9:45	0:15	**Digital Attendance (AM) + Welcome**	Trainer welcomes learners and administers the mandatory SSG digital attendance. Housekeeping and safety briefing.	Trainer-led · Demonstration	Slides 2–8
9:45 – 10:15	0:30	**Introductions, Outcomes and Course Outline**	Trainer and learner introductions. Learning outcomes, TSC alignment, course outline, lesson plan and assessment briefing are presented.	Trainer-led · Class discussion	Slides 9–20
10:15 – 11:15	1:00	**Topic 1: Power Query Fundamentals**	What Power Query is and the ETL loop it automates. Power Query versus worksheet formulas. The ribbon, the Editor's five areas, and Applied Steps as an audit trail. Data types and why they must be set first.	Trainer-led · Demonstration · Q&A	Slides 21–30
11:15 – 12:30	1:15	**Labs 01–03: Loading, Types, Connections and Fill**	Lab 01 — load a table and fix every data type. Lab 02 — connect to an external workbook and load it as Connection Only. Lab 03 — repair a ragged report with Fill Down and Fill Up.	Hands-on practical · Individual work	Slides 31–34
12:30 – 13:30	—	***LUNCH BREAK***	*Not counted as instructional time.*	—	—
13:30 – 13:40	0:10	**Digital Attendance (PM)**	Mandatory SSG digital attendance for the afternoon session.	Administration	—
13:40 – 15:10	1:30	**Topic 1: Shaping Columns**	Splitting columns by delimiter — into columns versus into rows, and why the choice matters. Conditional, custom and example-driven columns. Merging columns, sorting	Trainer-led · Demonstration · Q&A	Slides 34–36

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			and filtering inside the query.		
15:10 – 16:10	1:00	**Labs 04–06: Splitting, Adding and Filtering**	Lab 04 — split a name and a product list. Lab 05 — add conditional, custom and example-driven columns. Lab 06 — merge columns, then sort and filter at source.	Hands-on practical · Individual work	Slides 37–42
16:10 – 17:30	1:20	**Topic 2: Logic and Dates in M**	The M language versus Excel formulas — case sensitivity and the compulsory else. Writing IF logic by hand. Nesting conditions with and / or. Date.AddDays and the deadline pipeline.	Trainer-led · Demonstration · Worked example	Slides 45–49
17:30 – 18:30	1:00	**Labs 07–09: IF, Nested Logic and Deadlines**	Lab 07 — write IF logic in M. Lab 08 — nest IF with AND for a priority flag. Lab 09 — calculate due dates and ageing buckets. Day 1 recap.	Hands-on practical · Individual work	Slides 50–54

Total instructional time, Day 1: 8 hours 00 minutes.

Day 2 — Combining, modelling and advanced analysis with Power Query and Power Pivot

Time	Dur.	Session / Topic	Content and Activities	Instructional Method	Slide Ref
9:30 – 9:40	0:10	**Digital Attendance (AM) + Day 1 Recap**	Mandatory SSG digital attendance. Recap of Day 1 and Q&A.	Administration · Class discussion	—
9:40 – 11:00	1:20	**Topic 2: Reshaping and Aggregating**	Why wide is for humans and tall is for machines. The three unpivot commands and the trap in Unpivot Only Selected Columns. Pivot and the Don't Aggregate option. Append versus merge. The Group By feature.	Trainer-led · Demonstration · Q&A	Slides 54–57
11:00 – 12:00	1:00	**Labs 10–12: Unpivot, Pivot, Append and Group By**	Lab 10 — unpivot a wide monthly report into tidy rows. Lab 11 — pivot without aggregation and append two quarters. Lab 12 — summarise 400	Hands-on practical · Individual work	Slides 58–63

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			transactions with Group By.		
12:00 – 12:30	0:30	**Topic 2: The Power Pivot Data Model**	The division of labour between Power Query and Power Pivot. Star schema — fact and lookup tables. Relationships. Implicit versus explicit measures, and the four DAX measures.	Trainer-led · Demonstration	Slides 63–68
12:30 – 13:30	—	***LUNCH BREAK***	*Not counted as instructional time.*	—	—
13:30 – 13:40	0:10	**Digital Attendance (PM)**	Mandatory SSG digital attendance for the afternoon session.	Administration	—
13:40 – 14:20	0:40	**Lab 13: Build a Power Pivot Model with DAX**	Load three tables to the Data Model, draw the relationships in Diagram View, and write four explicit DAX measures, verifying each in a PivotTable.	Hands-on practical · Individual work	Slide 69
14:20 – 15:40	1:20	**Topic 3: Joins and Merging**	How a merge works and the step everyone forgets. All six join kinds and the rows each keeps. Choosing a join. Merge versus XLOOKUP. Merging on multiple fields and the row-count inflation trap.	Trainer-led · Demonstration · Q&A	Slides 72–77
15:40 – 16:40	1:00	**Labs 14–16: The Six Joins and Composite Keys**	Lab 14 — run all six joins and record the row counts. Lab 15 — replace XLOOKUP with a Left Outer merge. Lab 16 — merge on a composite key.	Hands-on practical · Individual work	Slides 78–81
16:40 – 17:40	1:00	**Topic 3 + Labs 17 –19: Advanced Matching**	Approximate match by binning and by banding table. Cross join and drill down. Fuzzy matching, the similarity threshold and transformation tables. Labs 17, 18 and 19.	Trainer-led · Hands-on practical	Slides 84–90
17:40 – 18:00	0:20	**Revision, Feedback and TRAQOM Survey**	Consolidation of the complete workflow, the common pitfalls, and the mandatory TRAQOM course-quality survey.	Class discussion · Administration	Slides 92–101
18:00 – 18:30	0:30	**Digital**	Mandatory	Administration ·	Slides 92–101

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		Attendance (Assessment) + Assessment Briefing	assessment digital attendance and the assessment briefing. The Written Assessment (1 hr) and Practical Performance (1 hr) follow.	Assessment	

Total instructional time, Day 2: 8 hours 00 minutes.

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6. Practical Activities (Labs)

The course is built around 19 hands-on labs. Each lab has its own folder under labs/ containing a starter Excel workbook with mock data and pre-built Power Query queries. Full step-by-step instructions for every lab are in the Learner Guide.

Lab	Title	Maps To	Starter Workbook
01	Load Sales Data and Fix Data Types	LO1 · A1 · K1, K4	lab-01-load-and-data-types.xlsx
02	Combine an External Workbook as a Connection Only	LO1 · A1 · K4	lab-02-external-connection.xlsx
03	Fill Down and Fill Up a Ragged Report	LO1 · A1 · K2	lab-03-fill-down-up.xlsx
04	Split Columns by Delimiter and into Rows	LO1 · A1 · K2	lab-04-split-columns.xlsx
05	Add Conditional, Custom and Example-Driven Columns	LO1 · A1 · K1, K2	lab-05-add-columns.xlsx
06	Merge Columns, Sort and Filter	LO1 · A1 · K2, K3	lab-06-merge-sort-filter.xlsx
07	Conditional Logic with IF in M	LO2 · A2 · K1, K2	lab-07-if-formulas.xlsx
08	Nest IF with AND for a Priority Flag	LO2 · A2 · K2	lab-08-nested-if-and.xlsx
09	Calculate Deadlines with Date.AddDays	LO2 · A2 · K1, K3	lab-09-adddays-deadline.xlsx
10	Unpivot a Wide Report into Tidy Rows	LO2 · A2 · K2, K3	lab-10-unpivot.xlsx
11	Pivot, Pivot Without Aggregation, and Append	LO2 · A2 · K2, K3	lab-11-pivot-append.xlsx
12	Summarise with Group By	LO2 · A2 · K1, K3	lab-12-group-by.xlsx
13	Build a Power Pivot Data Model with DAX Measures	LO2 · A2 · K3, K4	lab-13-power-pivot-model.xlsx
14	Walk Through All Six Join Kinds	LO3 · A3 · K2, K3	lab-14-six-joins.xlsx
15	Left Outer Join versus XLOOKUP	LO3 · A3 · K2, K3	lab-15-merge-vs-xlookup.xlsx
16	Merge on Multiple Fields	LO3 · A3 · K2	lab-16-multi-field-merge.xlsx
17	Approximate Match: Binning and Conditional Columns	LO3 · A3 · K2, K3	lab-17-approximate-match.xlsx
18	Cross Join and Drill Down to a Variable	LO3 · A3 · K1, K2	lab-18-cross-join-drilldown.xlsx
19	Fuzzy Matching and Transformation Tables	LO3 · A3 · K2, K3	lab-19-fuzzy-matching.xlsx

7. Assessment

Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.

Practical Performance (PP) — hands-on Power Query and Power Pivot tasks, 1 hour, open book.

A minimum of 75% attendance is required to be eligible for assessment and funding.

Assessment is open book: learners may refer to the slide deck, the Learner Guide and any approved course material, but may not use the internet, AI assistants, or discuss with other learners. Learners assessed Not Yet Competent are offered one re-assessment opportunity.

Criteria for funding

- A minimum attendance rate of 75%, based on the SSG digital attendance record.
- Completion of the assessment with an outcome of Competent.

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8. Training Resources

- Trainer slide deck (PPTX) and learner slide deck (PDF)
- Learner Guide (PDF) with step-by-step instructions for all 19 labs
- This Lesson Plan (PDF)
- 19 lab folders, each with a starter Excel workbook containing mock data and pre-built Power Query queries
- A laptop per learner with Microsoft Excel for Windows (Microsoft 365, 2021, 2019 or 2016), projector and whiteboard

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